# Diamond DA42 NG Twin Star (DA42)

2× Austro Engine AE300 turbo-diesel, 168 hp each (FADEC / ECU, single power lever per engine) · 4 seats · Typical DA42 NG (AE300, G1000) AFM procedures; speeds are NG figures and differ from the DA42 TDI (Thielert) – switch labels (ECU A/B voter, ECU TEST) vary slightly by serial.

## V-speeds & limits

| | |
|---|---|
| Vr | **76 KIAS (flaps UP) / 71 KIAS (flaps APP)** |
| Vx | **84 KIAS** |
| Vy | **90 KIAS** |
| Vmca | **68 KIAS** |
| Vyse | **85 KIAS** |
| Vxse | **82 KIAS** |
| Best glide | **88 KIAS (both props feathered, flaps UP)** |
| Vfe | **137 KIAS (APP) / 113 KIAS (LDG)** |
| Vle / Vlo | **194 KIAS extended / 156 KIAS retraction** |
| Va | **125 KIAS** |
| Vno | **155 KIAS** |
| Vne | **194 KIAS** |
| Vs / Vso | **68 / 62 KIAS** |
| Vref | **86 KIAS (flaps LDG)** |

## Normal procedures

### Before Start

- Preflight inspection — **COMPLETE (fuel drains, oil & coolant levels, gearbox oil, tow bar removed)**
- Passenger briefing — **COMPLETE (belts, canopy / rear door, exits, no smoking)**
- Seats, belts, harnesses — **ADJUST & LOCK**
- Front canopy — **CLOSED & LOCKED (or cooling gap for taxi)**
- Rear door — **CLOSED & LOCKED**
- Parking brake — **SET**
- Gear selector — **DOWN**
- Power levers — **IDLE**
- Fuel selectors — **NORMAL (both)**
- Fuel transfer pumps — **OFF**
- ECU A/B (voter) switches — **AUTO (both)**
- Alternators L / R — **OFF**
- Avionic master — **OFF**
- Electric master — **ON**
- Annunciators / G1000 — **CHECK (GEAR three green, DOOR message off)**
- Fuel quantity — **CHECK mains and aux vs visual check**
- Flaps — **CHECK operation, set UP**
- Strobes / position lights — **ON**

### Engine Start

- Propeller area — **CLEAR (start LEFT engine first, then RIGHT)**
- Power lever — **IDLE**
- ENGINE MASTER — **ON (ECU self-test, glow plugs preheat)**
- L / R GLOW annunciation — **WAIT until extinguished**
- Starter — **ENGAGE (max 10 s, then 2 min cooling)**
- Oil pressure — **CHECK rising within 3 s, green within 30 s**
- RPM — **CHECK idle (approx 890 RPM) / engine stable**
- ALTERNATOR — **ON – check charging, no L/R ALTN FAIL**
- Repeat for other engine — **COMPLETE**
- Avionic master — **ON**
- Engine instruments — **CHECK (oil / coolant / gearbox temps rising)**
- Pitot heat — **TEST then OFF**

### Before Taxi

| | |
|---|---|
| Flight instruments / G1000 | **CHECK & SET (altimeter, HSI, CDI)** |
| Radios / transponder | **SET** |
| Trims (elevator / rudder) | **SET for takeoff** |
| Flaps | **UP (or APP for short field)** |
| Parking brake | **RELEASE** |
| Brakes | **CHECK immediately after moving** |

Nosewheel is free-castoring: steer with differential brakes and asymmetric power. Avoid riding the brakes.

| | |
|---|---|
| Turn coordinator / HSI | **CHECK during turns** |

### Before Takeoff

| | |
|---|---|
| Parking brake | **SET** |
| Flight controls | **FREE & CORRECT** |
| Trims | **SET takeoff** |
| Fuel selectors | **NORMAL (both)** |
| Fuel transfer pumps | **OFF** |
| Engine temperatures | **OIL min 50°C, COOLANT min 60°C for takeoff** |
| Ground check – LEFT engine: | **POWER LEVER IDLE** |
| ECU TEST L button | **PRESS & HOLD – RPM rises, propeller cycles to feather and back, ECU A FAIL / ECU B FAIL appear then clear** |
| ECU TEST L button | **RELEASE after test complete (no ECU caution remains)** |
| ECU A/B switch L | **B – engine runs normally; A – engine runs normally; back to AUTO** |
| Ground check – RIGHT engine: | **REPEAT ECU TEST and ECU A/B check** |
| Power levers | **IDLE (check RPM stable, no asymmetric indication)** |
| Engine instruments | **CHECK green, no caution messages** |
| Electrical | **CHECK voltage / both alternators charging** |
| Flaps | **UP or APP as planned** |
| Front canopy | **CLOSED & LOCKED (no gap)** |
| Rear door | **LOCKED (DOOR annunciation off)** |
| Pitot heat / lights | **AS REQUIRED** |
| Transponder | **ALT** |
| Takeoff briefing | **COMPLETE (Vr 76, Vyse 85, Vmca 68, engine failure plan, gear up at positive rate)** |

### After Takeoff / Climb

| | |
|---|---|
| Takeoff power | **CHECK 100% load / 2,300 RPM both engines** |
| Airspeed | **Vy 90 KIAS (obstacles: Vx 84 KIAS), never below Vyse 85 once gear is up** |
| Landing gear | **UP (positive rate, below 156 KIAS) – lights out** |
| Flaps | **UP (if APP used, above 85 KIAS)** |
| Climb power | **SET (92% load or as required; FADEC manages mixture / prop)** |
| Engine instruments | **CHECK (oil / coolant / gearbox temps green)** |

### Cruise

| | |
|---|---|
| Power levers | **SET (typ. 60–75% load, cruise speed 150–160 KTAS)** |
| Fuel quantity | **CHECK mains & aux, balance within limits** |
| Fuel transfer pumps | **ON when main tank below approx 2/3 – transfers aux to main, OFF when complete (L/R FUEL TRANSFER message)** |

Do not transfer with a nearly full main tank – surplus fuel vents overboard. Transfer stops when the aux tank is empty.

| | |
|---|---|
| Engine instruments / ECU | **CHECK (no ECU A / B caution)** |
| Electrical | **CHECK voltage / load** |

### Descent

| | |
|---|---|
| Altimeter | **SET** |
| Power levers | **REDUCE as required (no shock-cooling limits, FADEC controlled)** |
| Fuel selectors | **NORMAL** |
| Fuel transfer pumps | **OFF** |
| Approach briefing | **COMPLETE (ATIS, approach, speeds, single-engine plan)** |

### Before Landing

| | |
|---|---|
| Seats / belts | **SECURE** |
| Fuel selectors | **NORMAL** |
| Airspeed | **BELOW 194 KIAS (Vlo extension)** |
| Landing gear | **DOWN – three green, hydraulic pump off** |
| Flaps | **APP below 137 KIAS, LDG below 113 KIAS (when landing assured)** |
| Airspeed | **FINAL 86 KIAS (flaps LDG), short field 82 KIAS** |
| Landing light | **ON** |
| Gear | **RE-CHECK three green on final** |

### After Landing

| | |
|---|---|
| Flaps | **UP** |
| Pitot heat | **OFF** |
| Transponder | **STBY / as required** |
| Lights | **AS REQUIRED** |
| Front canopy | **COOLING GAP position permitted for taxi** |

### Shutdown

| | |
|---|---|
| Parking brake | **SET** |
| Power levers | **IDLE, 2 min cool-down (turbo)** |
| Avionic master | **OFF** |
| Alternators L / R | **OFF** |
| ENGINE MASTER L | **OFF** |
| ENGINE MASTER R | **OFF** |
| Electric master | **OFF (after propellers stop)** |
| Fuel selectors | **NORMAL (leave on)** |
| Canopy | **CLOSED, sun cover installed** |
| Pitot covers / chocks / tie-downs | **INSTALL** |

## Abnormal procedures

### ECU A or ECU B Failure

L/R ECU A FAIL or L/R ECU B FAIL caution

| | |
|---|---|
| ECU A/B switch (affected engine) | **CHECK AUTO – the ECU voter has switched to the serviceable unit** |
| Engine | **CHECK running normally (RPM, load)** |
| Power lever | **AVOID rapid movements** |

**▶ Both ECU A and ECU B caution on same engine?**

**Yes:**

| | |
|---|---|
| ECU A/B switch | **TRY A, then B manually** |
| Engine failure | **EXPECT – prepare one-engine-inoperative procedure** |
| Land | **AS SOON AS POSSIBLE** |

**No – single ECU:**

Deferred: One ECU failed – no redundancy on that engine, land as soon as practical. ..... **NOTED**

### Alternator Failure / Low Voltage

L/R ALTN FAIL caution or LOW VOLTS warning

| | |
|---|---|
| ALTERNATOR (affected side) | **OFF, then ON (reset once)** |

**▶ Alternator restored?**

**Yes:**

| | |
|---|---|
| Electrical load | **MONITOR** |

**No – one alternator:**

| | |
|---|---|
| ALTERNATOR (affected side) | **OFF** |
| Electrical load | **REDUCE to within single alternator capacity** |

Deferred: One alternator failed – single-source electrics, reduced load, land as soon as practical. ..... **NOTED**

**No – both failed / LOW VOLTS:**

| | |
|---|---|
| Both ALTERNATOR switches | **OFF** |
| Non-essential loads | **OFF (lights, pitot heat if not required, air conditioning)** |
| Battery | **MONITOR (approx 30 min endurance at reduced load – verify per AFM)** |

Engines keep running on ECU backup batteries. Extend gear early while battery power remains, or use emergency extension.

Deferred: On battery only – extend gear early (or free-fall), flaps once on final, land as soon as possible. ..... **NOTED**

## Low Oil Pressure / High Oil Temperature

L/R OIL PRES or L/R OIL TEMP caution

Oil pressure / temperature (affected engine) ..... **CHECK and cross-check**

Power lever (affected engine) ..... **REDUCE**

**▶ Pressure below red line or temperature rising above limit?**

**Yes:**

Engine ..... **SHUT DOWN & FEATHER (ENGINE MASTER OFF) before seizure**

Continue per One Engine Inoperative.

Deferred: One engine shut down – single-engine approach, Vyse 85 KIAS. ..... **NOTED**

**No – borderline:**

Power ..... **MINIMUM practical, monitor**

Deferred: Oil pressure / temperature abnormal – land as soon as practical, be ready to shut down the engine. ..... **NOTED**

## High Coolant / Gearbox Temperature

L/R COOLANT TEMP or L/R GBX TEMP caution

Power lever (affected engine) ..... **REDUCE**

Airspeed ..... **INCREASE (more cooling airflow)**

Coolant / gearbox temperature ..... **MONITOR**

**▶ Temperature returning to green?**

**Yes:**

Power ..... **KEEP REDUCED, monitor**

Deferred: Engine overheated – reduced power on one engine, maintenance inspection. ..... **NOTED**

**No:**

Engine ..... **SHUT DOWN & FEATHER (ENGINE MASTER OFF)**

Deferred: One engine shut down for overheat – single-engine approach, Vyse 85 KIAS. ..... **NOTED**

## Low Fuel / Fuel Pressure

L/R FUEL LOW or L/R FUEL PRESS caution

Fuel quantity ..... **CHECK main and aux (affected side)**

Fuel transfer pump (affected side) ..... **ON if aux fuel remains**

Fuel selector (affected engine) ..... **X-FEED from other main tank if imbalance / starvation**

**▶ Caution extinguished?**

**Yes:**

Fuel ..... **MONITOR, keep crossfeed only as needed**

Deferred: Fuel system abnormal – land as soon as practical. ..... **NOTED**

**No:**

Engine failure ..... **EXPECT – prepare one-engine-inoperative procedure**

Land ..... **AS SOON AS POSSIBLE**

## Gear Fails to Retract / Gear Unsafe Light

Airspeed ..... **BELOW 156 KIAS**

Gear selector ..... **CHECK UP, recycle DOWN then UP once**

GEAR circuit breaker ..... **CHECK IN**

**▶ Gear retracted and lights out?**

**Yes:**

Flight ..... **CONTINUE, monitor hydraulic pump cycling**

**No:**

Gear selector ..... **DOWN – three green**

Airspeed ..... **LIMIT to 194 KIAS, plan fuel for gear-down drag**

Deferred: Gear down and locked – return for landing, do not retract, increased fuel burn. ..... **NOTED**

## Canopy or Rear Door Open in Flight

DOOR caution or canopy / door unlatched

Airspeed ..... **REDUCE below 137 KIAS (Vfe APP)**

Canopy / door ..... **DO NOT attempt to close in flight**

The canopy stays in the cooling gap position; the rear door opens a few inches. Expect noise and buffet; the aircraft remains controllable.

Loose objects / belts ..... **SECURE**

Deferred: Canopy / door open – land as soon as practical, keep speed below 137 KIAS. ..... **NOTED**

### PFD / AHRS / ADC Failure

| | |
|---|---|
| Display backup (red button) | **PRESS – reversionary mode on MFD** |
| Standby instruments | **USE (airspeed, attitude, altimeter, compass)** |
| Autopilot | **OFF if AHRS failed** |
| PFD / AHRS / ADC circuit breakers | **CHECK** |

**▶ Flight in IMC?**

**Yes:**

| | |
|---|---|
| ATC | **ADVISE, request radar assistance / nearest VMC** |
| Deferred: Primary flight display failed – partial panel, request vectors, avoid IMC. | **NOTED** |

**No:**

| | |
|---|---|
| Deferred: PFD / AHRS failed – standby instruments only. | **NOTED** |

### Inadvertent Icing Encounter

**LAND AS SOON AS PRACTICAL**

| | |
|---|---|
| Pitot heat | **ON** |
| De-ice system (if fitted) | **ON (HIGH / MAX), check fluid** |
| Exit icing conditions | **TURN BACK / CHANGE ALTITUDE** |
| Power | **INCREASE to keep propellers clear** |
| Autopilot | **OFF, monitor control feel** |
| Landing | **FLAPS UP or APP, approach 95–100 KIAS, gear down late** |
| Deferred: Residual airframe ice – flaps-up / APP landing, approach 95–100 KIAS, longer runway. | **NOTED** |

## Emergency procedures

### Engine Failure During Takeoff Roll

| | |
|---|---|
| Power levers (memory) | **BOTH IDLE** |
| Brakes (memory) | **APPLY, maintain direction with rudder / differential brake** |
| ENGINE MASTERS | **OFF (both) when stopped** |
| Fuel selectors | **OFF** |
| Electric master | **OFF** |

### Engine Failure after Takeoff (gear down / low altitude)

Airborne, insufficient runway remaining

**LAND IMMEDIATELY**

| | |
|---|---|
| Power levers (memory) | **BOTH MAX** |
| Airspeed (memory) | **85 KIAS (Vyse) – never below 68 KIAS (Vmca)** |
| Landing gear (memory) | **UP (if climb required)** |
| Flaps (memory) | **UP** |
| Failed engine (memory) | **IDENTIFY (dead foot – dead engine), VERIFY with power lever** |
| ENGINE MASTER (failed engine) (memory) | **OFF – propeller feathers automatically** |

Bank approx 5° into the operating engine, ball half out. If a climb at Vyse cannot be achieved or runway remains, land straight ahead: gear DOWN, flaps as required, both ENGINE MASTERs OFF before touchdown.

| | |
|---|---|
| Feathering | **CONFIRM (propeller stopped / low RPM); if not feathered expect reduced climb** |
| ALTERNATOR (failed engine) | **OFF** |
| Fuel selector (failed engine) | **OFF** |
| Return | **ONE-ENGINE-INOPERATIVE APPROACH** |

## Engine Failure in Flight / One Engine Inoperative

Loss of power on one engine with altitude available

**LAND AS SOON AS POSSIBLE**

- Power levers (memory) — **BOTH MAX (as required)**
- Airspeed (memory) — **85 KIAS (Vyse) or higher**
- Landing gear / flaps (memory) — **UP**
- Failed engine (memory) — **IDENTIFY (dead foot – dead engine), VERIFY with power lever**

**▶ Time and altitude for a restart attempt?**

**Yes – restart:**

- ECU A/B switch (failed engine) — **CHECK AUTO, try B then A**
- Fuel selector (failed engine) — **X-FEED (if fuel starvation suspected), fuel quantity check**
- Power lever (failed engine) — **IDLE**
- ENGINE MASTER (failed engine) — **OFF, then ON (ECU resets, glow)**
- Starter — **ENGAGE if propeller stopped (below 15,000 ft)**

**▶ Engine restarted?**

**Yes:**

- Power — **INCREASE slowly, monitor temperatures**
- ALTERNATOR — **ON**
- Deferred: Engine restarted in flight – land as soon as possible, maintenance inspection. — **NOTED**

**No:**

Secure the engine as below.

**No – secure engine:**

Securing the failed engine:

- Power lever (failed engine) — **IDLE**
- ENGINE MASTER (failed engine) — **OFF – propeller feathers automatically**
- Feathering — **CONFIRM (propeller stopped / low RPM)**
- ALTERNATOR (failed engine) — **OFF**
- Fuel selector (failed engine) — **OFF (or X-FEED to operating engine for balance)**
- Electrical load — **REDUCE (single alternator)**
- Bank / rudder — **5° into operating engine, ball half out, rudder trim**
- ATC — **DECLARE, request nearest suitable airport**

Deferred: One engine inoperative – single-engine approach, Vyse 85 KIAS, gear and full flap only when landing assured, go-around not possible below approx 4

## One-Engine-Inoperative Approach & Go-Around

One engine secured and feathered

- Fuel selector (operating engine) — **NORMAL or X-FEED as required**
- Airspeed — **85 KIAS (Vyse) minimum until landing assured**
- Flaps — **APP when established on final (below 137 KIAS)**
- Landing gear — **DOWN – three green when landing assured (approx 500 ft AGL)**
- Flaps — **LDG only when landing assured (below 113 KIAS)**
- Airspeed — **FINAL 86 KIAS, reduce power lever gradually**
- Rudder trim — **NEUTRAL before touchdown**

**▶ Go-around required?**

**Yes – above approx 400 ft AGL:**

- Power lever (operating engine) — **MAX**
- Flaps — **APP, then UP above 85 KIAS**
- Landing gear — **UP (positive rate)**
- Airspeed — **85 KIAS (Vyse), bank 5° into operating engine**

Single-engine go-around below approx 400 ft AGL or with gear down and full flaps may be impossible – commit to land.

**No – continue:**

- Landing roll — **EXPECT asymmetric braking / yaw, keep straight with rudder and brake**

## Engine Fire in Flight

**LAND IMMEDIATELY**

- Power lever (affected engine) (memory) — **IDLE**
- ENGINE MASTER (affected engine) (memory) — **OFF – propeller feathers**
- Fuel selector (affected engine) (memory) — **OFF**
- ALTERNATOR (affected engine) (memory) — **OFF**
- Cabin heat / defrost (memory) — **OFF**
- Airspeed — **INCREASE if fire persists (sideslip away from fire)**
- Landing — **ONE-ENGINE-INOPERATIVE APPROACH, nearest suitable field (do not restart)**

## Engine Fire During Start

- ENGINE MASTER (affected engine) (memory) ..... **OFF**
- Fuel selector (affected engine) (memory) ..... **OFF**
- ENGINE MASTER (other engine) ..... **OFF**
- Electric master ..... **OFF**
- Fire extinguisher ..... **OBTAIN**
- Aircraft ..... **EVACUATE**

## Cabin / Electrical Fire

**LAND AS SOON AS POSSIBLE**

- Electric master (memory) ..... **OFF**
- Avionic master (memory) ..... **OFF**
- Cabin heat / fresh air ..... **CLOSE**
- Fire extinguisher ..... **ACTIVATE**

Engines keep running: each ECU has its own backup battery when the electric master is off. Gear and flaps are electric/hydraulic – plan an emergency (free-fall) gear extension.

### ▶ Fire out and electrical power required?

**Yes:**

- Circuit breakers ..... **CHECK – do not reset tripped breakers**
- Electric master ..... **ON**
- Avionics / loads ..... **ON one at a time, check for smoke**
- Deferred: Electrical fire – suspect circuits isolated, affected systems unavailable. ..... **NOTED**

**No:**

Continue with electrics off. Standby instruments only, no flaps, gear by emergency extension.

- Deferred: All electrics off – emergency gear extension, flaps-up landing (approx 95 KIAS), no radios. ..... **NOTED**

## Emergency Descent

- Power levers (memory) ..... **IDLE**
- Flaps (memory) ..... **UP**
- Landing gear (memory) ..... **DOWN (below 194 KIAS) to steepen descent**
- Airspeed (memory) ..... **194 KIAS smooth air / 155 KIAS turbulence**

## Emergency Gear Extension (free-fall) / Gear-Up Landing

Gear does not extend normally or hydraulic / electrical failure

- Airspeed ..... **REDUCE below 156 KIAS**
- GEAR (hydraulic pump) circuit breaker ..... **PULL**
- Gear selector ..... **DOWN**
- Emergency gear extension handle ..... **PULL fully and HOLD until three green (gear free-falls, approx 30 s)**
- Landing gear ..... **CHECK three green (test light bulbs if not)**

### ▶ Three green?

**Yes:**

- Yaw / sideslip ..... **ASSIST lock if needed**
- Deferred: Gear extended by emergency system – do not retract, hydraulic pump CB left out, normal landing, maintenance inspection. ..... **NOTED**

**No – gear-up / unsafe landing:**

- Airport ..... **SELECT longest paved runway, emergency services alerted**
- Fuel ..... **BURN DOWN to minimum practical, fuel transfer pumps OFF**
- Seats / belts / loose objects ..... **SECURE**
- Flaps ..... **LDG**
- Approach ..... **NORMAL 86 KIAS, minimum rate of descent at touchdown**
- Front canopy ..... **UNLOCK to cooling gap position before touchdown**
- ENGINE MASTERS ..... **OFF just before touchdown (propellers feather)**
- Fuel selectors ..... **OFF**
- Electric master ..... **OFF after touchdown**
- Aircraft ..... **EVACUATE when stopped**